

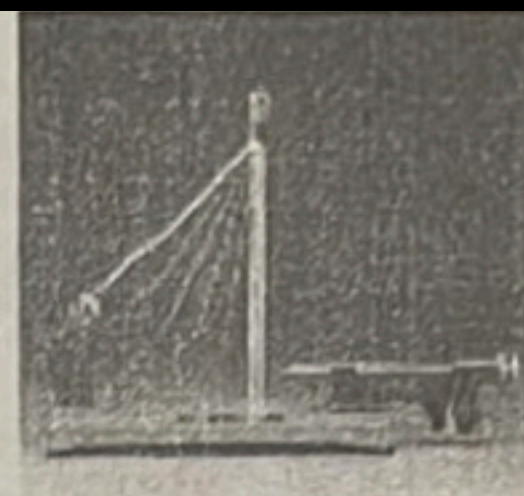
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Investigating the Ballistic Pendulum App

Name

A ballistic pendulum is a device for measuring a bullet's momentum, from which it is possible to calculate the velocity that a bullet was fired with. Although ballistic pendulums have been largely rendered obsolete by modern chronographs which measure the bullet's velocity directly, they are a great way to demonstrate conservation of energy and momentum.



Part A:

Let's model the problem . . . there are two important conservation concepts to divide "situation" into . . .

Concept # 1:

Conservation of mechanical energy.

Perform the initial algebra work to generate an equation . . . (3 marks)

$$E_{m_i} = E_{m_f}$$

$$E_{g_i} + E_{k_i} = E_{g_f} + E_{k_f}$$

$$\frac{1}{2}mv^2 = mgh_f$$

$$\frac{1}{2}v_i^2 = gh_f$$

$$v_i = \sqrt{2gh_f}$$

v_i is speed of both masses
(pendulum + ball)

Concept # 2:

Conservation of momentum

Perform the initial algebra work to generate an equation . . . (3 marks)

m_1 = mass of ball

m_2 = mass of pendulum

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$\frac{m_1 v_{1i}}{m_1} = \frac{v_f (m_1 + m_2)}{m_1}$$

$$v_{1i} = \frac{v_f (m_1 + m_2)}{m_1}$$

They stick so $v_{1f} = v_{2f}$ velocity from before

And the velocity of the pendulum at the start is zero so

$$v_{2i} = 0.00 \text{ m/s}$$



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Let's put both equations together to calculate the initial velocity of the "bullet" ... Your equation should show $V_i =$

$$(0.069 \text{ kg} + 0.270 \text{ kg})$$

(2 marks)

V_f of Concept 2 is V_i of Concept 1. ... final velocity of the collision is the initial velocity of both masses.

$$V_i = \sqrt{2gh_f} = V_f$$

$$h_f = 0.135 \text{ m} - 0.05 \text{ m}$$

$$h_f = 0.085 \text{ m}$$

$$m_1 = 0.069 \text{ kg}$$

$$m_2 = 0.270 \text{ kg}$$

$$g = 9.81 \text{ m/s}^2$$

$$V_{ii} = (\sqrt{(9.81 \text{ m/s}^2)(0.085 \text{ m})})$$

$$0.069 \text{ kg}$$

$$V_{ii} = 6.3 \text{ m/s}$$

$$V_{ii} = \frac{V_f (m_1 + m_2)}{m_1}$$

$$V_{ii} = \frac{(\sqrt{2gh_f})(m_1 + m_2)}{m_1}$$

Initial velocity of ball in class: 6.34 m/s ✓

Part B:

Using the initial velocity formula you derived above, determine the initial velocity of the ball if the mass of the ball was 350g, mass of the pendulum was 500g and the height was 8.5 cm. (2 marks)

$$m_1 = \text{mass of ball} = 0.350 \text{ kg}$$

$$m_2 = \text{mass of pendulum} = 0.500 \text{ kg}$$

$$h_f = 0.085 \text{ m}$$

$$V_{ii} = \frac{(\sqrt{2gh_f})(m_1 + m_2)}{m_1}$$

$$= \frac{(\sqrt{2(9.81 \text{ m/s}^2)(0.085 \text{ m})})(0.350 \text{ kg} + 0.500 \text{ kg})}{0.350 \text{ kg}}$$

$$V_{ii} = 3.14 \text{ m/s} ✓$$



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